A Wearable Asynchronous Brain-Computer Interface Based on EEG-EOG Signals With Fewer Channels

Li Hu, Junbiao Zhu, Sicong Chen, Yajun Zhou, Zhiqing Song, and Yuanqing Li, *Fellow, IEEE*

***Abstract*—Objective: Brain-computer interfaces (BCIs) have tremendous application potential in communication, mechatronic control and rehabilitation. However, existing BCI systems are bulky, expensive and require laborious preparation before use. This study proposes a practical and user-friendly BCI system without compromising performance. Methods: A hybrid asynchronous BCI system was developed based on an elaborately designed wearable electroencephalography (EEG) amplifier that is compact, easy to use and offers a high signal-to-noise ratio (SNR). The wearable BCI system can detect P300 signals by processing EEG signals from three channels and operates asynchronously by integrating blink detection. Result: The wearable EEG amplifier obtains high quality EEG signals and introduces preprocessing capabilities to BCI systems. The wearable BCI system achieves an average accuracy of 94.03±4.65%, an average information transfer rate (ITR) of 31.42±7.39 bits/min and an average false-positive rate (FPR) of 1.78%. Conclusion: The experimental results demonstrate the feasibility and practicality of the developed wearable EEG amplifier and BCI system. Significance: Wearable asynchronous BCI systems with fewer channels are possible, indicating that BCI applications can be transferred from the laboratory to real-world scenarios.**

***Index Terms*—Brain-computer interface (BCI), fewer channels, wearable, asynchronous, P300 signals.**

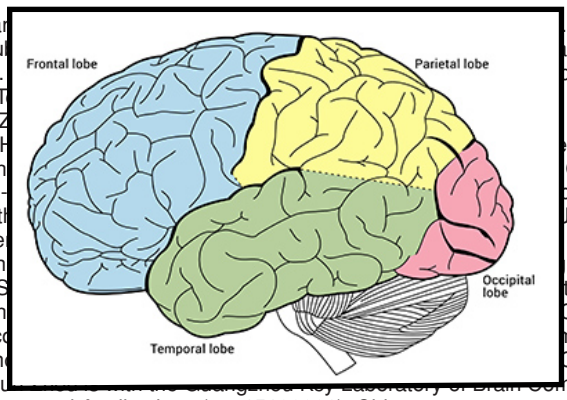
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Yaju… Interface and Applications (no. 15180006), China.



Yuanqing Li is with the School of Automation Science and Engineering, South China University of Technology, Guangzhou 510640, China, and also with the Guangzhou Key Laboratory of Brain Computer Interface and Applications (no. 15180006), Guangzhou 510640, China (e-mail: auyqli@scut.edu.cn).

Digital Object Identifier 10.1109/TBME.2023.3308371

I. Introduction

BRAIN-COMPUTER interfaces (BCIs) create communication pathways between brains and devices by processing and classifying brain signals [1]. These systems can help people suffering from motor disorders or facing certain situations by improving their communication and environmental control capabilities. Brain waves can be recorded by positron emission tomography (PET), functional magnetic resonance imaging (fMRI), magnetoencephalography (MEG), near-infrared spectroscopy (NIRS), and electroencephalography (EEG) [2]. When features are extracted from brain activity, computers can decode the user's subjective intentions and the brain's objective state through artificial intelligence (AI) and edge computing algorithms [3].

Over the last decade, significant progress has been made in the experimental paradigms, feature extraction, and decoding algorithms used in BCI systems. However, the hardware and user comfort of BCI systems have long been neglected [4]. Excessive cables and enormous size limit the use of BCI systems outside the laboratory. Moreover, unmodifiable data acquisition interfaces reduce programming flexibility [5]. In addition, due to poor EEG caps and tedious operation processes, most existing BCI systems are not acceptable for healthy individuals [6]. Recently, wearable devices have been increasingly used in real-life applications due to improvements in their usability and comfort [7]. NIRS and EEG are the most commonly used brain imaging modalities in wearable BCI system for capturing brain activity. [2]. NIRS measures fluctuations in cerebral hemodynamics, however, the light is easily scattered and absorbed by brain tissues. Moreover, subject movement and heart rate changes can affect the oxygenation level in different brain regions [8]. In contrast, EEG directly reflects of neural activity with high temporal resolution and minimal environmental constraints, making it highly suitable for portable recording of externally stimulated evoked potentials in the brain.

The P300 potential recorded by EEG signals, which can be detected in parietal and occipital regions at approximately 300 ms after transient stimulation is a potentially practical BCI paradigm [10]. Building an ideal wearable P300 BCI system requires the development of a low-cost, wireless, small size, low power, high signal quality EEG acquisition device with a minimal number of channels. However, meeting all these requirements remains a

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Spatial resolution describes an imaging system's ability to distinguish fine details or identify two separate objects as distinct entities

challenging task due to the low spatial resolution and signal-to-noise ratio (SNR) of EEG signals. For example, some wireless EEG devices, such as those proposed in [5], [11], have SNRs as good as those obtained by wired EEG amplifier. However, without an appropriate detection algorithm, an excessive number of channels are required to detect P300 signals. Commercial EEG headsets, such as the NeuroSky [12] and Emotiv Epoc [13] EEG devices, are primarily used to detect steady-state visual evoked potential (SSVEP) signals and endogenous EEG signals for attention, meditation, motor imagery (MI) and emotion recognition [14], [15], [16], [17], [18]. Few attempts have been made to develop a P300 BCI system based on commercial EEG devices with fewer channels. Wearable ear-EEG BCIs are easy to use and comfortable. However, ear-EEG signals are mainly used to capture temporal lobe brain activity, and the signal strength is weak. Furthermore, for practical applications, the performance of ear-EEG BCI needs to be improved [9]. Some studies have attempted to improve the accuracy and comfort of P300 BCI systems by using dynamic montages, such as [19], but the variable electrode locations still require the user to wear EEG caps, which limits the comfort and practicality of such BCI systems.

Synchronous P300 BCI systems generate outputs at specific time intervals [20]. To address this issue, many studies have attempted to implement asynchronous P300 BCI systems. some systems are based on statistical analysis of P300 signal features, such as [23], while other systems are based on multiple signals [21], such as combining P300 potential with event-related desynchronization (ERD) [25], electrooculography (EOG) [26], [27], MI [9], NIRS [43] and SSVEP [24]. EOG signals reflect potential fluctuations caused by eye movements (e.g., fixations, saccades, pursuits and blinks). In contrast to EEG signals, EOG signals are easier to acquire and provide good classification accuracy for switch control and event selection. In addition, EOG systems have a fast response times and stable performance across different subjects. Therefore, the combination of P300 and EOG signals may enable the development of hybrid BCI systems that can perform complex tasks and effectively switch between control and idle states.

In this work, we report a wearable BCI system with a good user experience that uses fewer channels to detect P300 signals and operates asynchronously by integrating blink detection. The proposed BCI system includes a self-developed front-end wearable EEG device and a back-end host system. The optimized circuits design and preprocessing module minimize the ambient electromagnetic and electrostatic noise of the EEG device. The synchronization module allows the EEG device to simultaneously record EEG signals, EOG signals, time markers, electrode impedance and electrode shedding. The graphical user interface (GUI) of the host system contains sixteen buttons that flicker according to the "odd-ball" paradigm to evoke P300 potentials. A support vector machine (SVM) model is applied to classify the P300 signal, and the amplitude variation rate of the EOG waveform is used to detect blinks. The results of P300 copy spelling test and dialing experiments demonstrate that the proposed BCI system can be utilized by users in real-life situations.

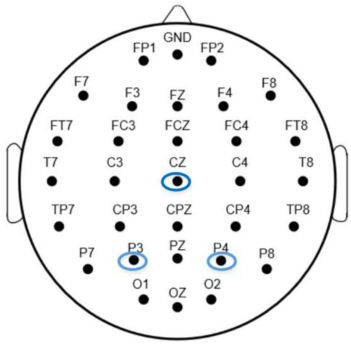


Fig. 1. EEG channels configuration (blue solid circles).

The remainder of this article is organized as follows: Section II describes the materials and methods, including the participants, the montages, the wearable EEG amplifier, the data acquisition process, the graphical user interface (GUI), the experimental paradigm and the P300 signals detection, blink detection and decision making algorithms; Section III presents the experimental setup and the detailed experimental results; Section IV evaluates and discusses the system performance; and Section V concludes the article and provides an outlook for future research.

II. Materials and Methods

A. Participants

Ten healthy native Chinese individuals (3 females and 7 males, aged 22 to 40 years) were recruited for this pilot study. All the subjects were right-handed and had normal or corrected-to-normal vision. None of the subjects had a relevant history of psychiatric or neurological disorders or other cerebral diseases. This work was approved by the Guangdong Work Injury Rehabilitation Hospital (approval number: AF/SC-07/2023.01). All subjects provided written informed consent prior to participation and comprehended the purpose and procedure of the experiments conducted in this study. Five subjects had previous experience with BCIs, whereas the other subjects had no exposure to BCIs.

B. Data Acquisition

Montages: As illustrated in Fig. 1, electrodes were placed at specific sites on the scalp according to the standard positions in the 10–20 system [23] to record EEG/EOG signals with a shared reference electrode. Three electrodes (Cz, P3, and P4) in the parietal and occipital regions were selected for EEG data acquisition. The electrodes placed on the forehead (GND) and the right mastoid (A2) were utilized as the ground and reference, respectiverly. We modified the interface of the EEG cap (32-channel Quik-Cap, Neuroscan, Inc. AUS) develop the wearable and wired EEG device In experiment I. In experiment II, the electrodes were placed directly on the scalp and secured with a custom-made head ring (Fig. 2(a)). The ground and reference electrodes were placed on both sides of the temples. The electrode placed over the user's eye was used to record EOG

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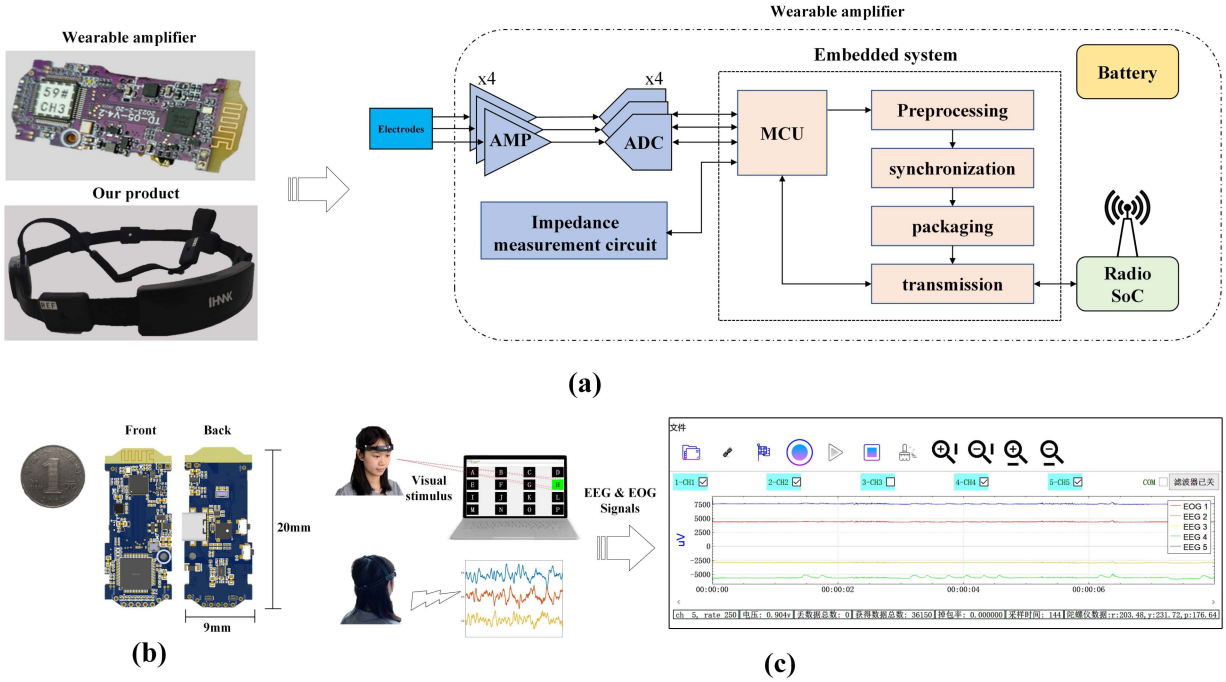


Fig. 2. (a) Schematic of the wearable EEG amplifier. The left subplot shows the unpackaged (top) and packaged (bottom) versions of our wearable EEG amplifier. The right subplot shows the hardware and embedded system implementation, AMP: amplifier chips, ADC: analog-to-digital (A/D) converters, MCU: microcontroller unit, and SoC: system-on-chip. (b) The size of the wearable EEG amplifier. The subplot shows the 3-D printed circuit board (PCB) of the wearable EEG amplifier and the coin is one yuan (CNY) for size comparison. (c) Experiment paradigm and the EEG and EOG signals visualization.

data. The impedance of all electrodes was maintained at less than 5 kΩ.

Wearable EEG amplifier: The wearable EEG amplifier consists of hardware circuits and an embedded system (Fig. 2(a)). The wearable EEG amplifier is 20 × 9 × 3 mm^3 in size and weighs 65 g (including the battery). High-precision and programmable gain instrumentation amplifier chips and high resolution and ultralow noise analog-to-digital (A/D) converters are used to obtain EEG and EOG signals from the electrodes. The impedance detection circuit can accurately track the contact impedance between the electrodes and the skin. If the impedance exceeds the designated threshold, the electrode is identified and removed. A radio system-on-chip (SoC) is employed for wireless data transmission. The system utilizes a battery as a power source, which helps to reduce interference from power ripples. The common mode rejection ratio (CMRR) of the amplifier is 110 dB, the input impedance of each EEG channel reaches up to 5 GΩ (at 20 Hz), the input-referred noise is less than 1 μVrms (0.5–100 Hz), and the average operating current is 20 mA.

All data preprocessing, synchronization, packaging and transmission are handled by an embedded system equipped with a powerful 32-bit ARM Cortex M4 microcontroller. The EEG and EOG signals are sampled at 250 Hz and filtered using an adaptive coherence model algorithm [27] to eliminate power-line interference. To ensure that the actions in the paradigm and the EEG/EOG signals are synchronized, the embedded system employs time markers to identify the locations of the EEG/EOG signals. The EEG and EOG signals, time markers and electrode impedance are combined into a data packet and numbered to prevent data misalignment. These data streams are wirelessly transferred to the host computer via Bluetooth Low Energy 5.0 (BLE 5.0) for real-time visualization (Fig. 2(c)), analysis and storage. The EEG and EOG data are recorded in the European Data Format (EDF).

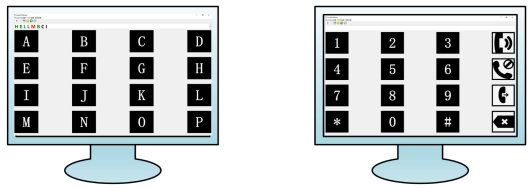


Fig. 3. GUI of our BCI system. A text box shows cue characters or selected characters at the top of the screen, and 16 buttons are displayed at the bottom. (a) GUI for experiment I. (b) GUI for experiment II.

C. Graphical User Interface (GUI) and Paradigm Design

As shown in Fig. 3, the GUI includes a text box at the top of the screen to present tips and messages and sixteen buttons corresponding to sixteen characters (Experiment I: 'A-P'; Experiment II: '0–9', '∗', '#', 'dial', 'suspend', 'redial', and 'delete') evenly spaced across the screen in a configuration of 4 rows and 4 columns. The GUI is displayed on a 13.5 inch screen with a resolution of 3000 × 2000 pixels, 260 pixels per inch (PPI) and a refresh rate of 60 Hz.

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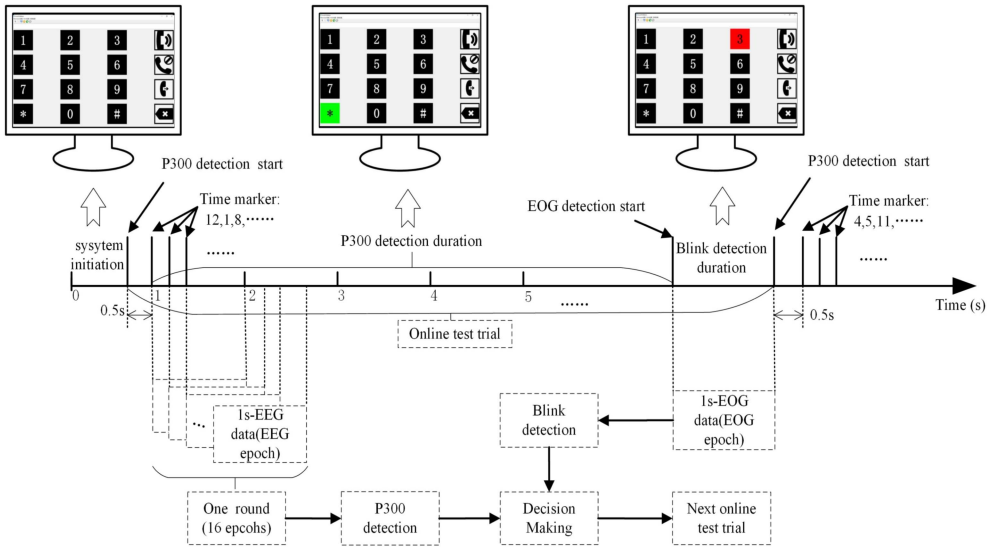


Fig. 4. Timeline and online test paradigm of the wearable asynchronous BCI system. The duration of the EEG/EOG epoch is 1 second and the interval between the start of the detection period and the first-time marker is 0.5 second. The three subplots at the top of the figure show the GUI during the system initiation, P300 detection, and EOG detection states. The green highlighted button indicates the visual cue for P300 stimulation and each stimulus have a time marker (the numbers 1 to 16 correspond to 16 characters.) to locate the start time. The red highlighted button represents the predicted results after P300 detection.

P300 paradigm: The P300 paradigm used in this study mainly follows the single display (SD) P300 paradigm [28]. Unlike the row-column (RC) paradigm [24], in the SD paradigm, only one character is highlighted at a time, and a complete flicker sequence in which all characters are pseudorandomly highlighted only once is called a round. In our study, each character was highlighted for 10 ms, followed by an interstimulus interval (ISI) of 100 ms without any highlighted characters. Thus, the duration of a round was 1.76 s (110 ms $\times$ 16). The P300 paradigm was divided into two blocks: the training block and the test block. The training block consisted of 16 trials, with each trial including 10 rounds. The subjects were asked to gaze without distraction at the button corresponding to the characters (target characters) presented in the text box and count the number of flashes silently. Note that each character was pseudorandomly selected to be a target only once. The EEG training data were saved for calibration purposes. For the test block, the subjects were asked to stare at the desired character and the program inferred the character based on the subject's EEG and EOG characteristics.

EOG paradigm: In the EOG training block, the cue button was illuminated for one second, followed by one second of darkness. The subject was instructed to blink once as comfortably as possible when the cue button is extinguished. The training EOG datasets were stored to calculate the subject-specific blink detection thresholds. During the EOG test block, the subject was instructed to blink in response to visual feedback (button turning red) to confirm that the output character is correct.

Fig. 4 illustrates the timeline and online testing paradigm of the wearable asynchronous BCI system. The online test process is divided into three stages: system initiation, P300 detection and blink detection. During the system initiation phase, a Bluetooth connection test, data stream test and parameter initiation were performed. After the system was initiated, the program began to extract P300 feature vectors from the EEG epochs for P300 detection. Depending on the classification score of the feature vectors, the program continued to acqurie more feature vectors or present more candidate character on the screen. For the blink detection process, the EOG epochs were used to identify blinks to determine whether the candidate character should be output before the start of a new trial. The comfort level of the paradigm is basically the same as the standard P300 paradigm, and blinking does not impose too much pressure on the user. The interval between adjacent P300 detection periods is referred to as an online test trial. The detailed EEG and EOG signal processing procedure is described below.

D. Signal Processing

P300 detection: The P300 detection scheme was used to identify P300 potentials in EEG signal (Fig. 5). The EEG training datasets consist of 2560 epochs (16 trials $\times$ 10 rounds $\times$ 16 buttons of flashes). For calibration, EEG time fragments (0–600 ms) were extracted from the EEG epochs, and the baseline was corrected by subtracting the mean amplitude of the EEG signal between -200 and 0 ms. Then, the signals were filtered a 10th order Butterworth filter with a bandwidth between 0.1 and 25 Hz and downsampled by a factor of 5. The EEG time fragments are denoted as $s^n_{k,r,j,\ l}$, where *n, k, r, j,* and *l* represent the *nth* button, the *kth* trial, the *r*th round (repeats), the *j*th channel and the number of samples ($l = 30$), respectively. Next, the feature vectors $\tilde{s}^n_{k,j*\ l}$ were calculated by concatenating the data vectors of all selected channels and determining the average value of all rounds. Finally, we obtained 16 target feature vectors and 240 nontarget feature vectors, each with a length of 90. In

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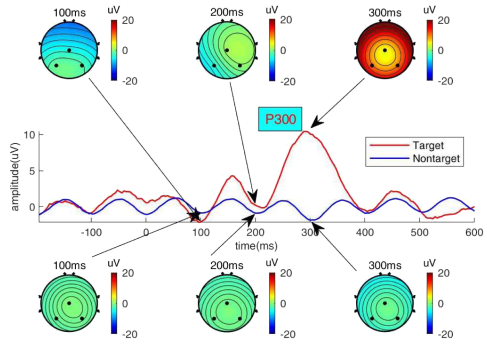


Fig. 5. P300 signals from a single channel of a specific subject. The red line is obtained by averaging all trial of target P300 signals and the blue line is obtained by averaging all trials of nontarget P300 signal. The brain topographies are obtained based on the target (top) and nontarget (bottom) P300 signals, respectively.

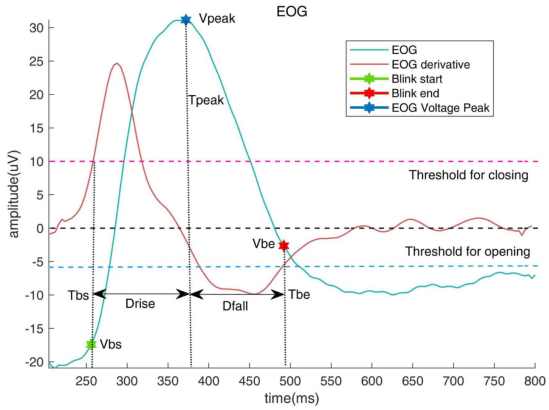


Fig. 6. Waveform of a blink EOG signal and the first derivative of the EOG signal.

the training process, the target feature vector and the nontarget feature vector were labelled 1 and -1, respectively.

The SVM model in the LIBSVM Toolbox was used to classify target and nontarget feature vectors with the radial basis function (RBF) as the kernel function. The trade-off parameter c and the kernel width g are two critical parameters that directly affect the generalizability of SVM model and the number of support vectors, respectively. To prevent overfitting and underfitting, we employed a grid search approach based on tenfold cross-validation to efficiently optimize the SVM model parameters. The optimized parameters were utilized to train the SVM model by incorporating feature vectors and labels. Finally, we emphasized that the SVM model must be saved after training to perform online testing.

During the online EEG signal testing procedure, the feature vectors $\tilde{S}$ were extracted from the EEG data following the calibration procedure. The SVM classification scores p were calculated as follows:

$$p^n = SVM\left(\frac{1}{N}\sum_{r=1}^{N}\tilde{S}_r^n\right) \tag{1}$$

where SVM represents the trained SVM model, and n and r are the nth flashing button and the rth round, respectively. N represents the total number of rounds needed to output one character. Next, the SVM scores were normalized in the range of [0, 1], as shown below:

$$\bar{p}^n = \frac{p^n - \min\limits_{s=1\cdots16} p^s}{\max\limits_{s=1\cdots16} p^s - \min\limits_{s=1\cdots16} p^s} \tag{2}$$

where $\bar{p}$ is the set of SVM scores after normalization. The candidate characters were determined based on the difference between the largest and second largest SVM scores, and the threshold $\bar{p}_{th}$ is chosen according to the specific needs of the application.

$$\bar{p}_{\max} - \bar{p}_{second} \geq \bar{p}_{th} \tag{3}$$

EOG detection: The response time of the human visual system is approximately 150 ms [29]. Thus, the EOG time fragments from 200 ms to 800 ms were extracted from the EOG epochs in the EOG training datasets. The baseline of the EOG time fragments was corrected by subtracting the mean amplitude of the EOG signal between 0 ms and 200 ms, and the signals were filtered by a 10th-order Butterworth filter with a bandwidth in the range of 0.1 to 10 Hz as well as a smooth filter. To identify blinks, the first-order derivative of the EOG signal was calculated (Fig. 6). The time of the signal peak was denoted as T_{peak}, while the intersections between the EOG derivative curve and the eyelid closing and opening thresholds indicated the times of blink onset and offset, which are denoted as T_{bs} and T_{be}, respectively. The eyelid closing and opening thresholds were empirically selected after analyzing datasets acquired during the training period. If T_{bs}, T_{be}, and T_{peak} were outside the range of 200 to 800 ms, they were replaced with the time of the boundary. The durations of eyelid closing (D_{rise}) and opening (D_{fall}) can be calculated as follows:

$$D_{rise} = T_{peak} - T_{bs} \tag{4}$$

$$D_{fall} = T_{be} - T_{peak} \tag{5}$$

The voltage values corresponding to the EOG signal peak, blink start and blink end are denoted as V_{peak}, $V_{bs}$and V_{be}, respectively. The rise (θ_{rise}) and fall (θ_{fall}) rates of the EOG signal amplitude are modeled as follows:

$$\theta_{rise} = \frac{V_{peak} - V_{bs}}{T_{rise}} \tag{6}$$

$$\theta_{fall} = \frac{V_{peak} - V_{be}}{T_{fall}} \tag{7}$$

The preprocessing and feature extraction in the online phase were identical to those in the training phase. The blinks lead to rapid increases and decreases in the amplitude of the EOG signal and persist for a certain duration. Additionally, due to distractions and participants not responding in a timely manner, imperfect blink waveforms were recorded frequently. Thus, we

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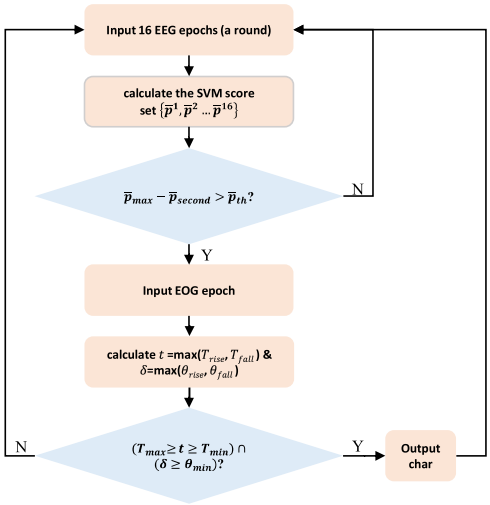


Fig. 7. Schematic of the decision-making procedure for one online test trial. $\{\bar{P}^1, \bar{P}^2, \ldots, \bar{P}^{16}\}$ represents the SVM scores corresponding to 16 characters.

set two criteria to locate the blinks as follows:

$$\text{blink} = \begin{cases} 1, & \text{if } (\mathrm{D}_{\max} \geq \max(D_{rise}, D_{fall}) \geq D_{\min}) \\ & \quad \cap (\max(\theta_{rise}, \theta_{fall}) \geq \theta_{\min}) \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

where $\mathrm{D}_{\max}$ and $D_{\min}$ are the maximum and minimum duration of opening and closing the eyelid during blinking in the training phase, and $\theta_{\min}$ is 60% of the mean signal voltage variation rate of closing or opening the eyelids during blinking in the training phase.

Decision making: The EEG signal processing procedure for P300 wave detection was applied for candidate selection, and the EOG signal processing procedure for blink detection was used to verify the results. The detailed decision procedure for the online test trials is shown in Fig. 7.

III. Experiments and Results

A. Copy-Spelling Test

In this experiment, we designed a copy-spelling task in which subjects spelled the same sequence of characters using wired (SynAmps2 equipment, Neuroscan, Inc. AUS) and wearable EEG amplifiers. The aim of the experiment was to evaluate the performance differences between the two amplifiers. The BCI system for the copy-spelling task operates in synchronous mode without an EOG signal. The experiment was conducted with unique software and invariable montages. Half of the subjects started the task with the wearable amplifier, and the other half started the task with the wired laboratory amplifier. The subjects were required to face screen at a distance of approximately 60 cm. To minimize the potential influence of fatigue and distraction, the task session was completed within 30 minutes. At the beginning of the experiment, a random sequence of 16 characters was displayed in the text box (Fig. 3(a)). The initial color of the characters in the sequence was black. If the character was spelt correctly, the color turned green. If the character was spelt incorrectly, the color turned red.

TABLE I
Online Testing Performance of all Subject Using Both BCI Systems

Subject		Accuracy (%)	Mean RT (S)	ITR (bits/min)
S1	wearable	100	6.44	34.94
	wired	93.75	5.87	20.93
S2	wearable	93.75	5.07	40.46
	wired	93.75	5.21	39.37
S3	wearable	93.75	4.87	42.12
	wired	100	6.24	38.26
S4	wearable	93.75	6.56	31.27
	wired	87.5	5.45	32.68
S5	wearable	87.5	4.89	36.42
	wired	100	5.74	41.81
S6	wearable	87.5	5.95	29.93
	wired	100	5.12	46.87
S7	wearable	93.75	5.83	35.18
	wired	93.75	5.09	40.30
S8	wearable	100	5.61	42.87
	wired	87.5	4.72	37.73
S9	wearable	87.5	6.26	28.45
	wired	87.5	5.73	31.08
S10	wearable	100	5.67	42.33
	wired	100	5.80	41.38
Mean ± SD	wearable	93.75±5.1	5.72±0.62	36.63±5.37
	wired	94.375±5.47	5.49±0.46	38.44±4.66

Red characters are incorrectly spelled characters, RT: response time for selecting a single target

Fig. 8 shows the offline classification accuracies (y-axis), which were calculated by 10-fold cross-validation with respect to the character flash times (repeats) (x-axis). The green line with the yellow dot represents the wearable amplifier and the blue line with the red dot represents the wired amplifier. Importantly, both amplifiers can achieve satisfactory classification accuracies under offline conditions.

To evaluate the online performance of both EEG amplifiers, the accuracy, mean response time (RT) for a single target selection and information transfer rate (ITR) were calculated (Table I). The ITR, which corresponds to the total information received by the BCI system per unit of time is widely used to assess the performance of BCI systems. The ITR (bits per minute) is calculated as follows [24]:

$$ITR = 60 \times \left(\log_2 M + P\log_2 P + (1-P)\log_2\left(\frac{1-P}{M-1}\right)\right)/T \tag{9}$$

where M is the number of characters, P is the accuracy of the BCI system, and T is the mean response time needed to select a character. Note that the preparation time was not included in the calculation of the ITR in this experiment.

On average, the wireless and wired amplifiers achieved relatively high accuracies (93.75%±5.1% and 94.375%±5.47%) and ITRs (36.63±5.37 bits/min and 38.44±4.66 bits/min), respectively.

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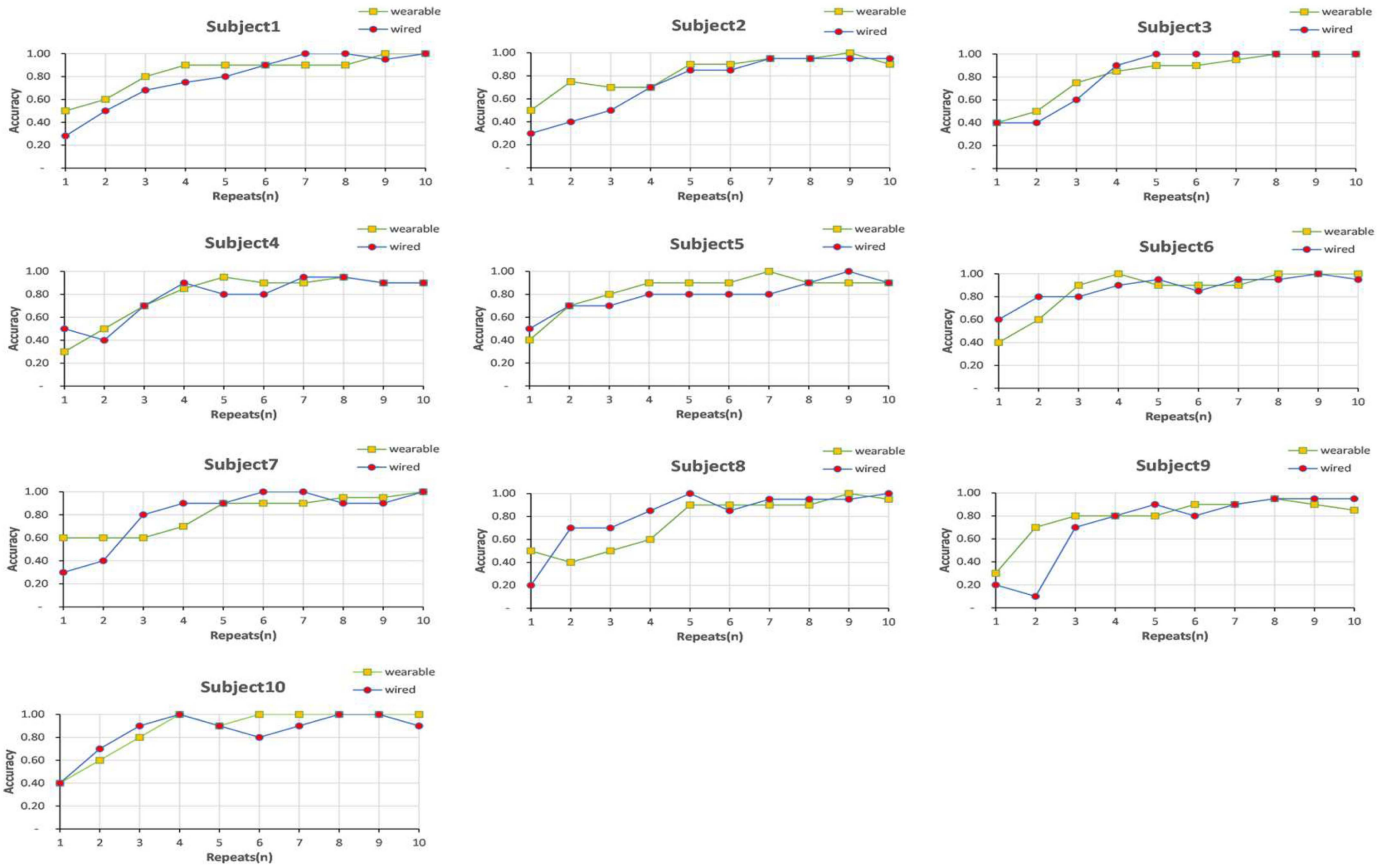


Fig. 8. Offline classification accuracy of wearable system (green line with yellow dot) and wired system (blue line with red dot) for copy-spelling.

B. Dialing Experiment With a Wearable Asynchronous BCI System

In the second experiment, we attempted to evaluate the performance of the wearable asynchronous hybrid BCI system in a dialing task. A telephone dialing platform was developed for this experiment: A network server was set up in a mobile phone using termux (an Android terminal emulator and Linux environment application), and the computer was connected to the mobile phone via Secure Shell Hibernate (SSH). The 4 × 4 button matrix used in our program acted as a virtual telephone keyboard (Fig. 3(b)).

During the experiment, subjects were asked to dial their own telephone number (eleven numbers) twice. The users input their telephone number by choosing the corresponding character, deleted incorrect numbers by choosing the delete button, redialed the previous phone number by choosing the redial button, stopped the phone call by choosing the suspend button and called the current phone number by choosing the dial button. Then, the subjects remained in the idle state for 10 minutes to assess the false-positive rate (FPR).

Table II illustrates the results of the telephone number dialing test. The accuracy, mean response time to select a single target, FPR and ITR were recorded. The FPR is a fatal evaluation indicator for asynchronous BCI systems. The FPR is defined as follows [30]:

$$FPR = \frac{FP}{TN + FP} \tag{10}$$

TABLE II
RESULTS OF THE PHONE NUMBER INPUT TEST

Subject	Mission Success	Accuracy (%)	FPR (%)	Mean RT (S)	ITR (bits/min)
S1	Yes	96.15	0	6.38	33.99
S2	Yes	90.32	3.22	7.17	26.47
S3	Yes	100	0	6.25	38.4
S4	Yes	96.43	7.14	5.6	42.76
S5	Yes	92.86	0	7.42	26.98
S6	Yes	90	0	7.78	24.22
S7	Yes	93.1	3.44	5.13	39.39
S8	Yes	96.15	0	6.54	33.16
S9	Yes	85.29	0	8.65	19.58
S10	Yes	100	4	8.22	29.20
Mean ± SD		94.03±4.65	1.78±2.25	6.91±1.13	31.42±7.39

The characters for one call consist of eleven digits and one dial button, RT: response time for selecting a single target

where *FP* is the number of false positive decisions and *TN* is the number of true negative decisions during the idle state of the experiment. In this study, the idle state (10 minutes) was divided into several online test trials (Fig. 4). At the end of each online test trial, if no character was outputted, the trial was counted as a *TN*, and if a character was outputted, the trial was counted as a *FP*. Notably, all subjects completed the task and show good performance using our wearable asynchronous BCI system. The

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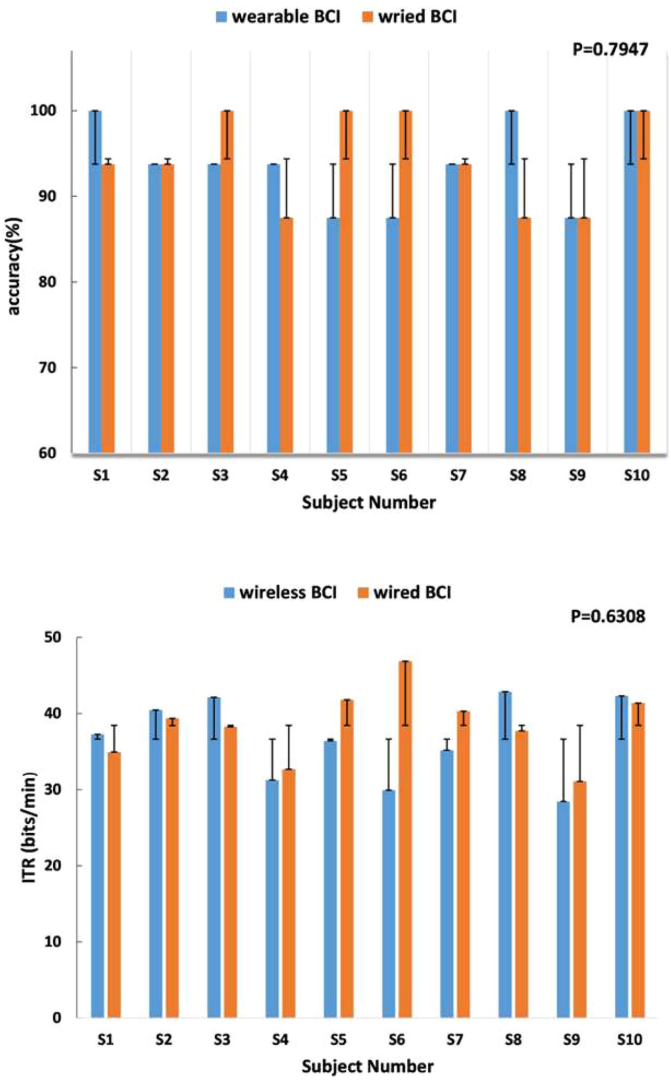


Fig. 9. Online classification accuracy and ITR results showing the standard deviation between the wearable and wired BCI systems with respect to the number of subjects.

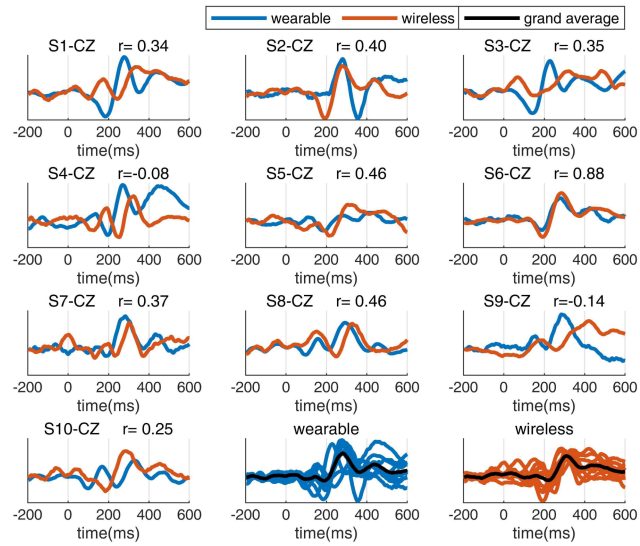


Fig. 10. P300 waveforms of each subject were obtained from channel Cz of the two EEG amplifiers and the P300 waveforms were derived by averaging EEG signals of all 16 trials. The last two subplots display the P300 waveforms of all participants recorded by the wearable and wired EEG amplifiers. The black line represents the grand average of the P300 waveforms across all participants. The correlation coefficient for the grand average P300 waveform is 0.7677 (r = 0.7677).

results show a high accuracy (94.03% ± 4.65%), relatively high ITR (31.42 bits/min ± 7.39 bits/min), short response time (6.91 s ± 1.13 s) in the control state and a low FPR (1.78% ± 2.25%) in the idle state.

IV. Discussion

The motivation of this work is to devolpe a lightweight and practical BCI system. To achieve this, we developed asynchronous BCI systems based on wearable EEG amplifiers and attempted to minimize the number of channels in the system. We believe that many laboratory BCI applications can be implemented using wearable BCI systems. The following discussion is divided into four parts. First, we present the performance of our wearable EEG amplifier. Second, we discuss P300 detection with fewer channels. Third, we discuss the performance and significance of wearable asynchronous BCI systems. Finally, we present the limitations of our system and provide ideas for future work.

To the best of our knowledge, P300 is currently the most reliable and stable signal for use in BCI systems [5], [25]. The results of the offline test (Fig. 8) and the online test (Table I) show that both BCI systems achieved satisfactory classification accuracies and ITRs. To compare the performance of the amplifiers, we calculated the average online classification accuracy and ITR with respect to the subjects (Fig. 9). Paired t tests were performed between the classification accuracy and ITR of the two BCI systems, yielding corresponding p values of 0.7947 and 0.6308. These p values indicated that there was no significant difference between the average online classification accuracies and average online ITRs between the two BCI systems. Furthermore, Fig. 10 illustrates the P300 waveforms collected from the Cz channels of all subjects using the two EEG amplifiers, and the correlation coefficients between the P300 waveforms were calculated. The results revealed that six subjects (s1, s3, s4, s7, s9, and s10, $|r| < 0.4$) exhibited weak correlations, three subjects (s2, s5, and s8, $0.4 < |r| < 0.7$) exhibited moderate correlations, and one subject (s6, $0.7 < |r|$) exhibited a strong correlation. The average correlation score across subjects was 0.329, revealing that the two amplifiers captured the cross-time variability of the P300 signals. The grand average P300 waveforms of all subjects from both BCI systems were surprisingly consistent with a correlation coefficient of 0.7677. This correlation score indicates that P300 signals have good reliability and repeatability. In conclusion, the wearable EEG amplifier has comparable accuracy, ITR, and P300 waveforms to the wired amplifiers. Therefore, we believe that both amplifiers are equally suitable for P300 detection.

Remarkably, our wearable asynchronous BCI system requires only three EEG signal channels and one EOG channel. The use

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TABLE III
COMPARISON BETWEEN OUR SYSTEM AND SEVERAL PREVIOUS WORK ON BCI SYSTEM

Reference	Amplifier mode	Work mode	Signals	Channels	Accuracy (%)	FPR (%)	ITR (bits/min)
[24]	wired & purchase	async	P300+EOG	7	93.6	2.6*	35.4
[25]	wired & purchase	async	P300+EOG	4	85*	-	18*
[32]	wired & purchase	async	P300+ SSVEP	11	91.33	-	47.14
[30]	wired & purchase	async	P300	16	93.71	5.15	43.15
[33]	wired & purchase	async	P300	16	94.4	-	-
[34]	wired & purchase	async	P300	8	87.46	1.73	11.19
[35]	wired & purchase	async	P300	8	95.75	0.7	31*
[20]	wired & purchase	async	P300+MI	18	92.6	4.5*	51.34*
[40]	wearable & purchase	sync	P300	`14	91.4	-	-
Proposed system	wearable & self-developed	async	P300+EOG	4	94.03	1.78	31.42

– denotes not mentioned in this study, * denotes the value was calculated according to data reported in the original article. Async: asynchronous, sync: synchronous.

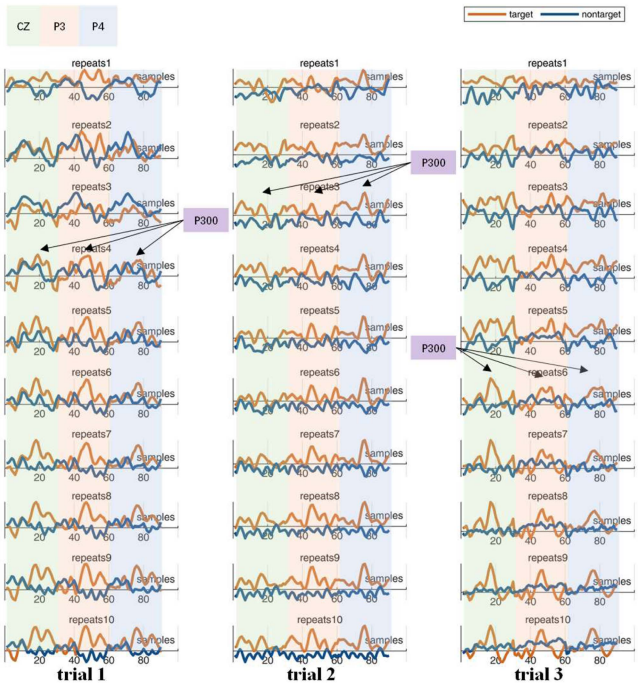


Fig. 11. Target (orange line) and non-target feature (blue line) vectors corresponding to various number of flashes from three trials. the target and non-target feature vectors are obtained by concatenating channels (CZ (light green), P3(light red), P4(light blue)). The P300 peaks begins to be clearly visible when the number of repeats was four, three and six respectively.

of fewer channels significantly reduces the system preparation time, increasing the ease of operation and the user friendliness of the system, which is more suitable for use in daily life. In our study, the feature vectors were obtained by concatenating the signals obtained from three channels (CZ, P3, and P4). The use of repetitive visual stimuli gradually enhances the P300 signal waveform. We found that the target feature vectors showed clear P300 peaks when the number of repeats was four, three and six (Fig. 11). This suggests that feature vectors with fewer channel signals can represent the differences between target and nontarget feature vectors well. visual evoked potentials are neuronal activity in the parietal and occipital areas of the cerebral cortex that are related to visual stimuli, and the "standard set" used for P300 detection is described in [31], which indicated that the components from Cz to Oz and P7 to P8 accounted for the largest portion of the variance.

Recently, many novel asynchronous BCI systems have been developed to address the disadvantages of traditional BCI systems, which have greatly improved people's lives [10], [20], [24]. Table II shows the performance of our asynchronous BCI system. All participants successfully completed the task, and the BCI system achieved satisfactory classification accuracy, FPR and ITR. Although the ITR is lower than that achieved in the copy-spelling test, the classification accuracy is higher. The results show that the proposed BCI systems can operate in asynchronous mode without reducing performance. Table III illustrates the performance of the proposed BCI system and existing BCI systems. The proposed BCI system has fewer channels than the other BCI systems. The classification accuracy of the proposed BCI system is close to the third-best score achieved among the existing BCI systems. Moreover, our BCI system is based on a wearable amplifier, which improves user friendliness and enriches the potential application scenarios of the BCI system. Based on the performance and advantages of our system, we suggest that the proposed BCI system has excellent application potential and promising market prospects.

Although our BCI system shows acceptable performance, the present study has three main limitations. First, visual stimuli cannot be directly presented on mobile phones. The intensity of the visual stimulus is limited by the screen of the mobile phone and compact buttons in the GUI can visually distract users' attention [31], [36]. Second, the performance and comfort of dry electrodes are compromised in hairy areas [37], [38]. Some specially shaped electrodes, such as comb-like electrodes, are uncomfortable for use. Third, wireless transmission delays, movement and vibration can affect the accuracy and ITR of BCI systems [14], [38]. Further research on BCI systems is needed to develop innovative rational paradigms, further improve the user experience and enhance the robustness of the system.

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V. Conclusion

In this study, we proposed a wearable asynchronous BCI system based on EEG-EOG signals with fewer channels. The experimental results demonstrate that the performance of our wearable EEG amplifier is comparable to that of a wired EEG amplifier. Furthermore, our asynchronous wearable BCI system achieves an average online accuracy of 94.03±4.65%, an average FPR of 1.78% and an average ITR of 31.42 bits/min ±7.39 bits/min. These results highlight the practical advantages of combining BCI systems with mobile devices. In future research, we will introduce transfer learning to simplify the calibration process and explore novel materials to enhance user comfort. More importantly, we will extend the wearable BCI system to more natural daily life applications for healthy and paralyzed people.

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